

SHUBHAN BHATIA

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EDUCATION

Ludwig-Maximilians-Universität München

2025 - Present

Master of Science in Astrophysics

Grade: 1.75 (German scale, 1.0 = best, 4.0 = pass)

New York University Abu Dhabi

2020 - 2024

Bachelor of Science in Physics

Cumulative GPA: 3.86/4.00

Founders' Day Award Recipient

PUBLICATIONS

S. Bhatia, L. Kimmig, R.-S. Remus, B. Seidel, K. Dolag, M. Kluge, S. Brough, How Old Are You Really? Validating the ICL+BCG Fraction as a Dynamical Age Tracer of Galaxy Clusters in Projection, in preparation, to be submitted to A&A.

M. Nori, **S. Bhatia**, A.V. Macciò, FUZZY-GASOLINE: **Cosmological hydrodynamical simulations of dwarf galaxy formation with Fuzzy Dark Matter** (arXiv:2411.09733).

RESEARCH EXPERIENCE

Graduate Research - Master's Thesis

Mar. 2026 - Present

Research Supervisor: Apl. Prof. Dr. Klaus Dolag

University Observatory Munich (USM), Ludwig-Maximilians-Universität München

- Building a pipeline to reconstruct cluster mass profiles from simulated X-ray gas density and temperature profiles under hydrostatic equilibrium.
- Estimating per-cluster, rather than population-averaged, hydrostatic mass bias by matching against a library of local-universe zoom-in analogs with known true masses.

Graduate Research - ICL+BCG Project

Jul. 2025 - Present

Research Supervisors: Dr. Lucas Kimmig and Dr. Rhea-Silvia Remus

University Observatory Munich (USM), Ludwig-Maximilians-Universität München

- Showed that the ICL+BCG fraction, a strong 3D tracer of cluster dynamical state, stays tightly correlated with its 2D projected counterpart across 42 local-universe cluster analogs, validating it as an observational proxy.
- Identified masking aperture choice, not projection angle, as the dominant term in the 2D ICL+BCG measurement's error budget. First-author manuscript to be submitted to A&A.

Research Assistant

Oct. 2024 - Nov. 2025

Research Supervisor: Prof. Joseph Gelfand

Center for Astrophysics and Space Science (CASS), New York University Abu Dhabi

- Built a UV variable-source catalog for the starburst galaxy IC 10 from a ~ 5 -year weekly Swift UVOT monitoring campaign, constructing and analyzing light curves for detected u -band sources.

Undergraduate Research - Bachelor's Thesis

Sep. 2022 - Feb. 2025

Research Supervisors: Prof. Andrea Macciò and Dr. Matteo Nori

Galaxy Formation Group, CASS, New York University Abu Dhabi

- Produced a suite of 30 hydrodynamical and dark-matter-only zoom-in simulations of dwarf galaxies (10^9 - $10^{11} M_\odot$) using the FUZZY-GASOLINE code on NYUAD's Jubail HPC cluster, part of the first

high-resolution hydrodynamical FDM galaxy-formation simulations.

- Found notably delayed star formation in lower-halo-mass galaxies under FDM, the clearest departure from the CDM case, with other stellar observables broadly consistent with NIHAO.

Research Assistant

May 2021 - Aug. 2021

Research Supervisor: Dr. Mohammad Ali-Dib

Center for Astro-Particle and Planetary Physics (CAP3), New York University Abu Dhabi

- Developed a supervised regression model to predict planetary orbital-stability limits in binary star systems, using training data from N-body simulations.

PROJECTS

sim-highline: Galaxy Scaling Relations from Cosmological Simulations (*beta*)

App | Code | DOI

- Open dataset and web app of about 1,600 curves from 20+ simulation suites (EAGLE, IllustrisTNG, SIMBA, COLIBRE and others); each curve carries its source, epoch, units, definitions and evidence tier, with scripted checks against published values and no interpolated epochs. Beta.

CONFERENCES AND WORKSHOPS

8th ICM Theory and Computation Workshop

Jul. 2026

Brno, Czech Republic – Poster: *Validating the ICL+BCG Fraction in Projection as a Dynamical Age Tracer of Galaxy Clusters Using the Lower Decks Zoom-in Simulations*

Cosmic Collisions

Jan. 2026

NYUAD, Abu Dhabi, United Arab Emirates – Attendee

RELEVANT COURSEWORK

MSc: Introduction to Advanced Astrophysics, Gravitational Dynamics, Hydrodynamics **BSc:** Astrophysics, General Relativity, Statistical Mechanics and Thermodynamics, Quantum Mechanics, Computational Physics

TECHNICAL SKILLS

Programming: Julia, Python, C++, C, Fortran

Scientific computing: NumPy, SciPy, Matplotlib, Pandas, Astropy, PyTorch

High-Performance Computing: Parallel computing (OpenMP, MPI), Job scheduling (SLURM)

Simulation & Analysis tools: Gasoline2, Gadget3, PyNbody, HEASoft, IRAF, ds9, Git, LaTeX

Languages: English (Native), Hindi (Fluent), German (Intermediate - B1), Arabic (Intermediate)

COMMUNITY ENGAGEMENT

Treasurer and Outreach Coordinator

2022 – 2024

Organized on- and off-campus astronomy outreach events, including an observatory visit, and secured recurring funding for *Spacebar*, NYUAD's student astronomy interest group.

Podcast Co-host – *Spaced Out*

2023 – 2024

Co-hosted and coordinated Season 3, handling guest selection and episode preparation for a public astronomy podcast.

REFERENCES

Apl. Prof. Dr. Klaus Dolag USM, LMU München

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Dr. Rhea-Silvia Remus USM, LMU München

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Dr. Lucas Kimmig University of Nottingham

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Prof. Andrea Macciò NYU Abu Dhabi

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